

Indian Institute Of Technology Gandhinagar



Assignment Report

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Department: Electrical Engineering

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Project

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Checkpoint

Design LDO with ideal opamp of 80 dB gain (Hint: Use VCVS as error amplifier and pass transistor). Create testbench for PSRR, Transient analysis, Loop Gain and phase, line and load regulation.

*Bonus [1 pt]: LDO with 5T OTA as error amplifier and PMOS as pass element. (Working stable LDO, may or may not match all the specs)

Specifications

- Input Voltage (V_{in}): 1.8 V
- Output Voltage (V_{out}): 1.5 V
- Load (output) Current (I_{load}): 50 mA (Nominal), 20 mA (light-load), 100 mA (full-load)
- C_{OUT} (OFF-CHIP) = 500 pF
- Compensation: There will be two methods for compensation. Please find the allotted method below.
 - Method 1: C_m (ON-CHIP) ≤ 15 pF (should be minimized) with nulling resistor in series with Miller capacitor.
 - Method 2: Buffer to be placed in between output of opamp and pass transistor.
- PSRR @ 100 kHz (dB): -40 dB
- Efficiency (η) $\geq 83\%$ (across load range)
- DC Error in V_{out} : $\leq \pm 0.1$ mV
- Load Regulation: $\leq 2\%$ variation from light-load to full-load
- Line Regulation: $\leq 2\%$ for $\pm 2.5\%$ variation in input voltage
- Transient Response: ≤ 150 mV overshoot/undershoot for a 20 mA to 50 mA load step (both up and down with edge time of 10 ns to be shown in the same plot) with settling time (output voltage within $\pm 2\%$ of the regulated value) less than 125 ns.
- Design to match performance specs at 0°C , 27°C (nominal), 60°C

Calculations

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} V_{ov}^2$$

Rearranging for $\frac{W}{L}$:

$$\frac{W}{L} = \frac{2I_D}{\mu_p C_{ox} V_{ov}^2}$$

Substituting the given values:

$$I_D = 100 \text{ mA} = 0.1 \text{ A}$$

$$V_{ov} = 0.3 \text{ V}$$

$$\mu_p C_{ox} = 65 \mu\text{A}/\text{V}^2 = 65 \times 10^{-6} \text{ A}/\text{V}^2$$

$$\frac{W}{L} = \frac{2 \times 0.1}{(65 \times 10^{-6}) \times (0.3)^2}$$

$$(0.3)^2 = 0.09$$

$$65 \times 10^{-6} \times 0.09 = 5.85 \times 10^{-6}$$

$$\frac{W}{L} = \frac{0.2}{5.85 \times 10^{-6}}$$

$$\boxed{\frac{W}{L} \approx 3.42 \times 10^4}$$

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Given:

$$L = 1 \mu\text{m}$$

$$W_{total} = \frac{W}{L} \times L$$

$$W_{total} = 3.42 \times 10^4 \times 1$$

$$W_{total} = 34188 \mu\text{m}$$

Choosing unit gate width:

$$W_{unit} = 100 \mu\text{m}$$

Multiplier:

$$M = \frac{W_{total}}{W_{unit}}$$

$$M = \frac{34188}{100}$$

$$\boxed{W = 100 \mu m, \quad L = 1 \mu m, \quad M = 341}$$

The feedback equation of the LDO is:

$$V_{out} = V_{ref} \left(1 + \frac{R_1}{R_2} \right)$$

Given:

$$V_{ref} = 0.8 V, \quad V_{out} = 1.5 V$$

$$\frac{R_1}{R_2} = \frac{V_{out}}{V_{ref}} - 1$$

$$\frac{R_1}{R_2} = \frac{1.5}{0.8} - 1$$

$$\frac{R_1}{R_2} = 0.875$$

Using quiescent current:

$$I_q = \frac{V_{out}}{R_1 + R_2}$$

$$12 \mu A = \frac{1.5}{R_1 + R_2}$$

$$R_1 + R_2 = \frac{1.5}{12 \times 10^{-6}}$$

$$R_1 + R_2 = 125 k\Omega$$

Since:

$$R_1 = 0.875 R_2$$

$$0.875 R_2 + R_2 = 125 k$$

$$1.875R_2 = 125k$$

$$R_2 = 66.7k\Omega$$

$$R_1 = 58.33k\Omega$$

$$R_1 = 58.33k\Omega, \quad R_2 = 66.7k\Omega$$

OTA DESIGN

Design of 5-Transistor OTA

Given Parameters

Given ICMR IS 0.8 to 1.4V

$$\mu_n C_{ox} = 300 \mu A/V^2$$

$$\mu_p C_{ox} = 65 \mu A/V^2$$

$$GBW = 10 \text{ MHz}$$

$$C_L = 1 \text{ pF}$$

$$V_{ov(n)} = 0.2 \text{ V}$$

$$V_{ov(p)} = 0.4 \text{ V}$$

Step 1: Calculate Required Transconductance

$$g_m = 2\pi \cdot GBW \cdot C_L$$

$$g_m = 2\pi \cdot 10^7 \cdot 10^{-12}$$

$$g_m = 62.83 \mu S$$

Step 2: Differential Pair (M1, M2)

Using:

$$g_m = \frac{2I_D}{V_{ov(n)}}$$

$$I_D = \frac{g_m V_{ov(n)}}{2}$$

$$I_D = \frac{62.83 \times 10^{-6} \cdot 0.2}{2}$$

$$I_D = 6.283 \mu A$$

Using saturation current equation:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov(n)}^2$$

$$6.283 \times 10^{-6} = \frac{1}{2} (300 \times 10^{-6}) \frac{W}{L} (0.2)^2$$

$$(0.2)^2 = 0.04$$

$$6.283 \times 10^{-6} = 6 \times 10^{-6} \frac{W}{L}$$

$$\boxed{\left(\frac{W}{L}\right)_{M1,M2} \approx 1.05}$$

Step 3: Tail Transistor (M5)

$$I_{tail} = 2I_D$$

$$I_{tail} = 12.566 \mu A$$

$$12.566 \times 10^{-6} = 6 \times 10^{-6} \frac{W}{L}$$

$$\boxed{\left(\frac{W}{L}\right)_{M5} \approx 2.1}$$

Step 4: PMOS Current Mirror Load (M3, M4)

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} V_{ov(p)}^2$$

$$6.283 \times 10^{-6} = \frac{1}{2}(65 \times 10^{-6})\frac{W}{L}(0.4)^2$$

$$(0.4)^2 = 0.16$$

$$6.283 \times 10^{-6} = 5.2 \times 10^{-6}\frac{W}{L}$$

$$\boxed{\left(\frac{W}{L}\right)_{M3,M4} \approx 1.21}$$

Final Summary

$$\left(\frac{W}{L}\right)_{M1,M2} \approx 1.05$$

$$\left(\frac{W}{L}\right)_{M5} \approx 2.1$$

$$\left(\frac{W}{L}\right)_{M3,M4} \approx 1.21$$

Power Consumption Calculation

Given:

$$V_{DD} = 1.8 \text{ V}$$

$$I_{total} = 100.2 \text{ mA} = 0.1002 \text{ A}$$

Power consumption is given by:

$$P = V_{DD} \times I_{total}$$

$$P = 1.8 \times 0.1002$$

$$P = 0.18036 \text{ W}$$

$$\boxed{P = 180.36 \text{ mW}}$$

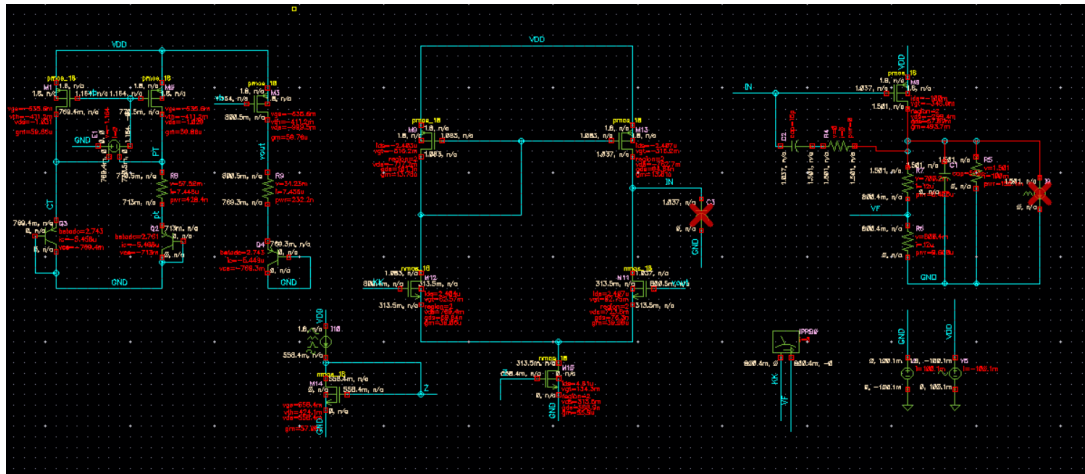


Figure 1: Complete Schematic of pmos LDO

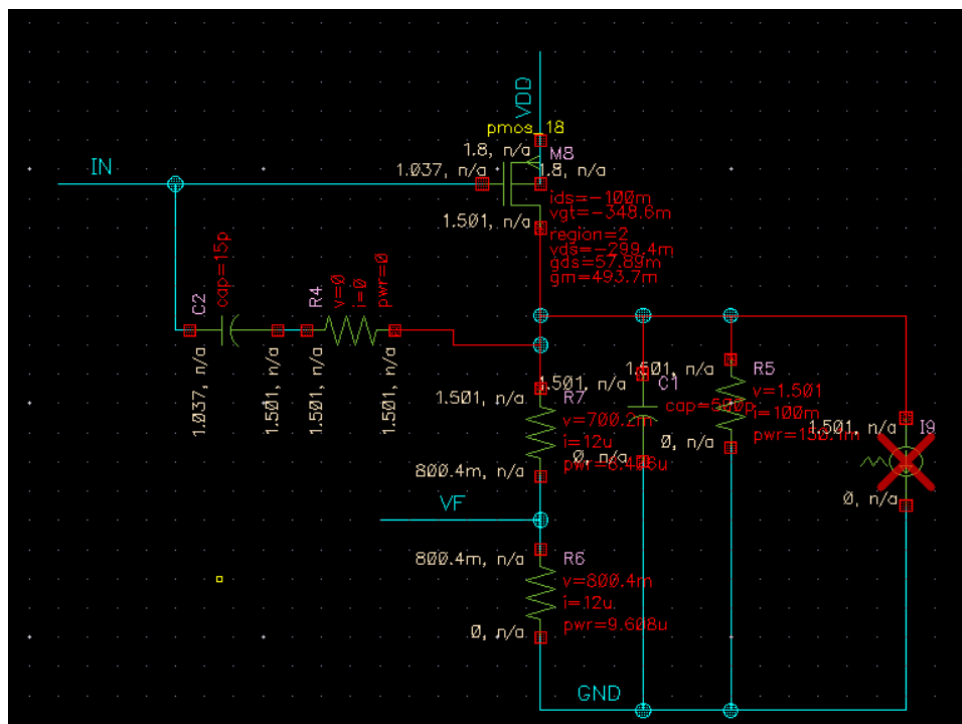


Figure 2: Schematic of pmos LDO

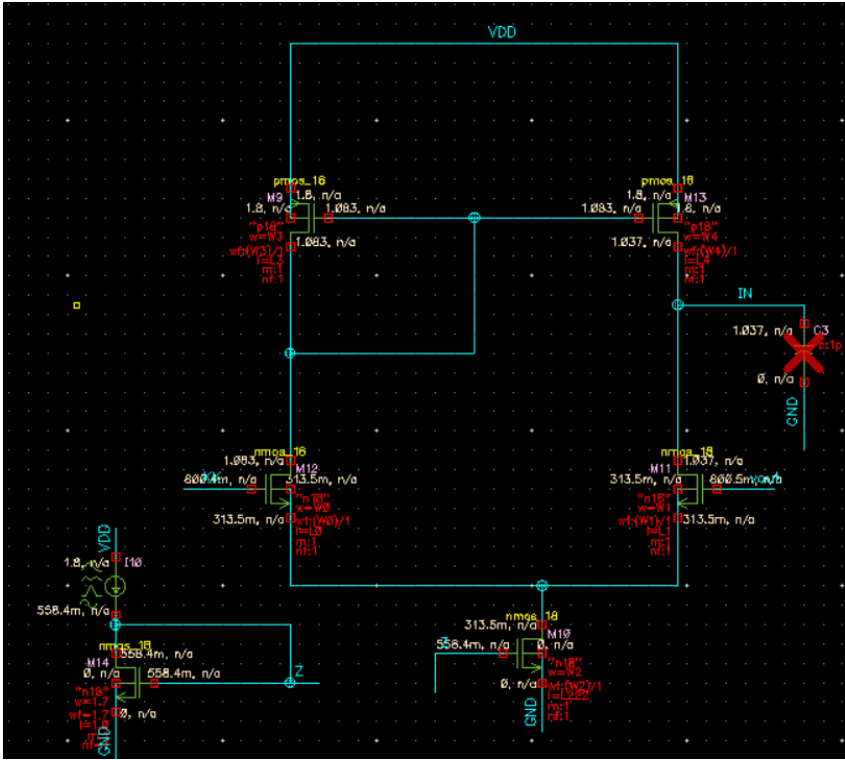


Figure 3: Schematic of transconducatace Amplifier

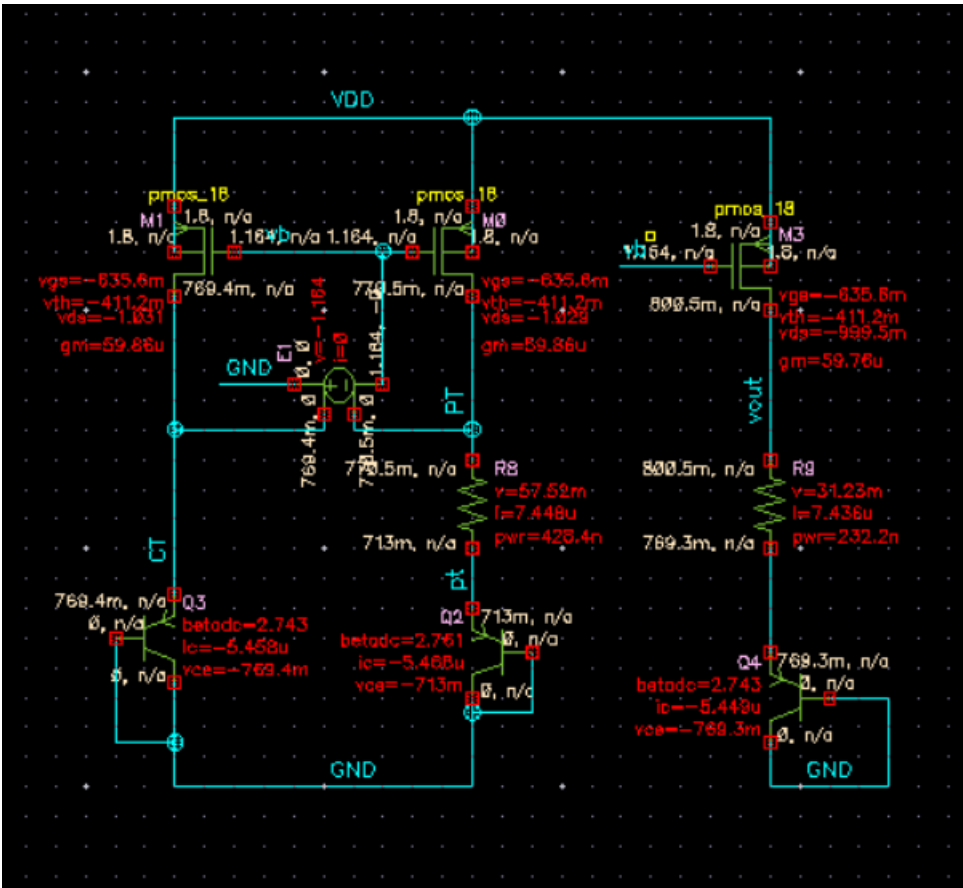


Figure 4: Schematic of BGR circuit Dessign

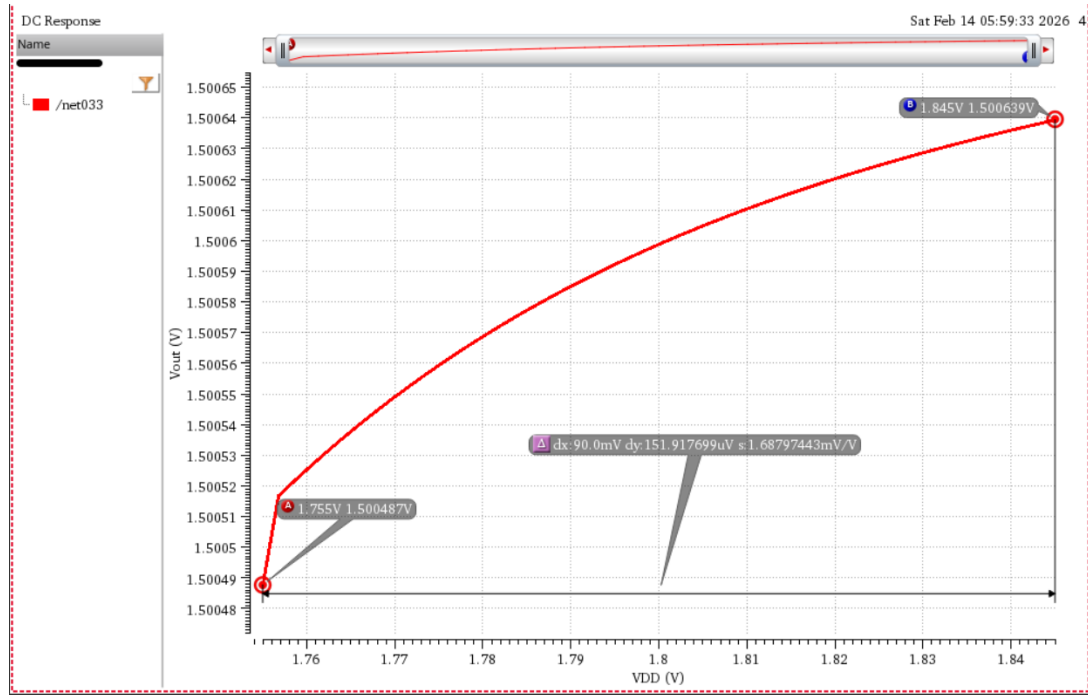


Figure 5: Complete Schematic of pmos LDO

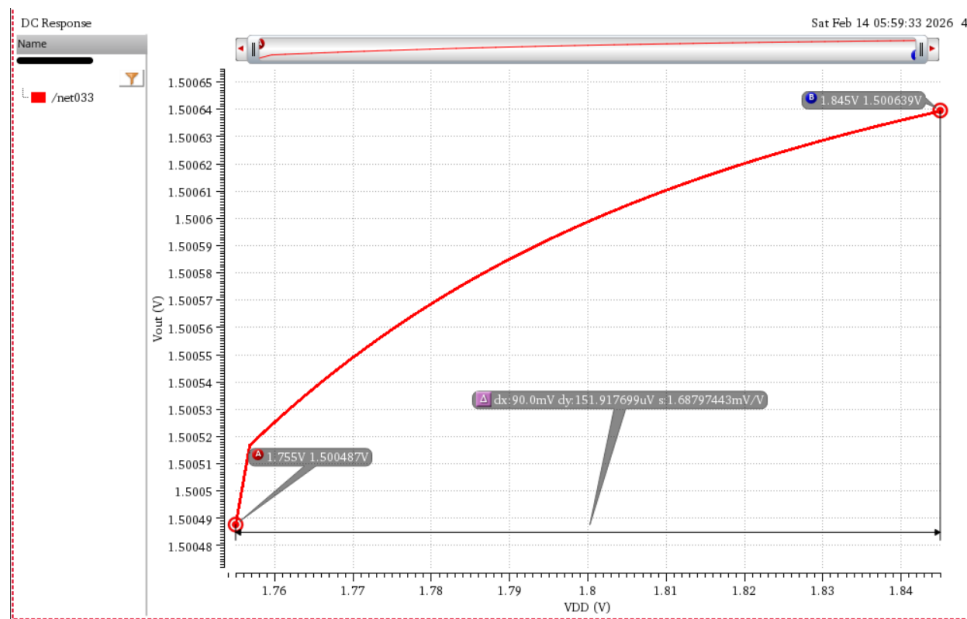


Figure 6: Line Regulation of pmos LDO

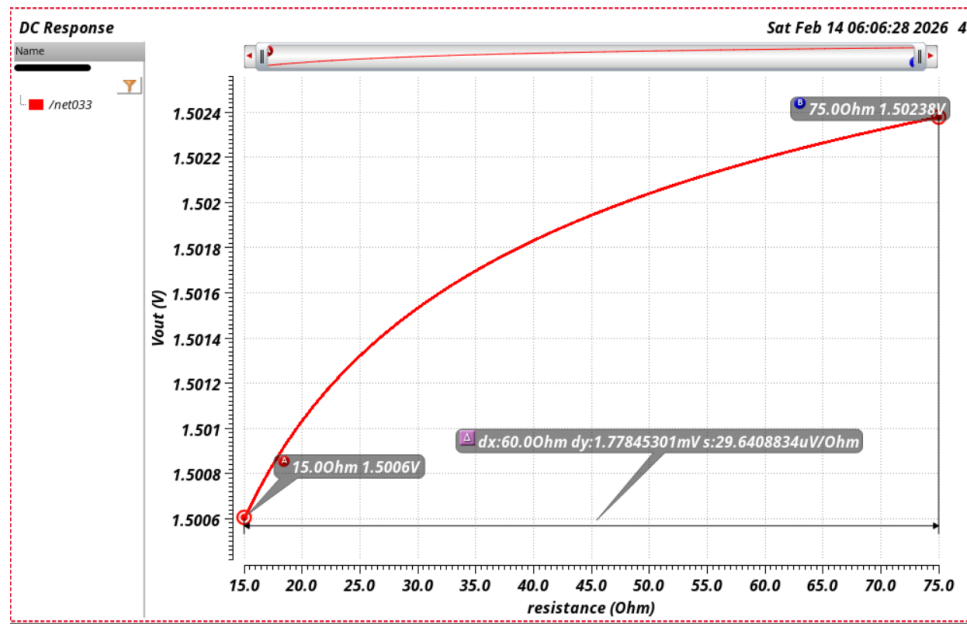


Figure 7: Load Regulation of pmos LDO

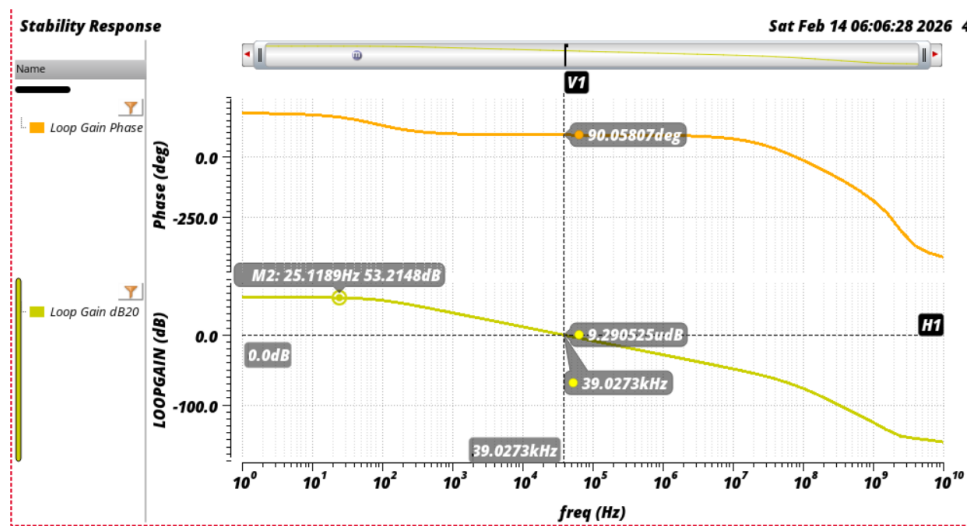


Figure 8: loop gain of pmos LDO

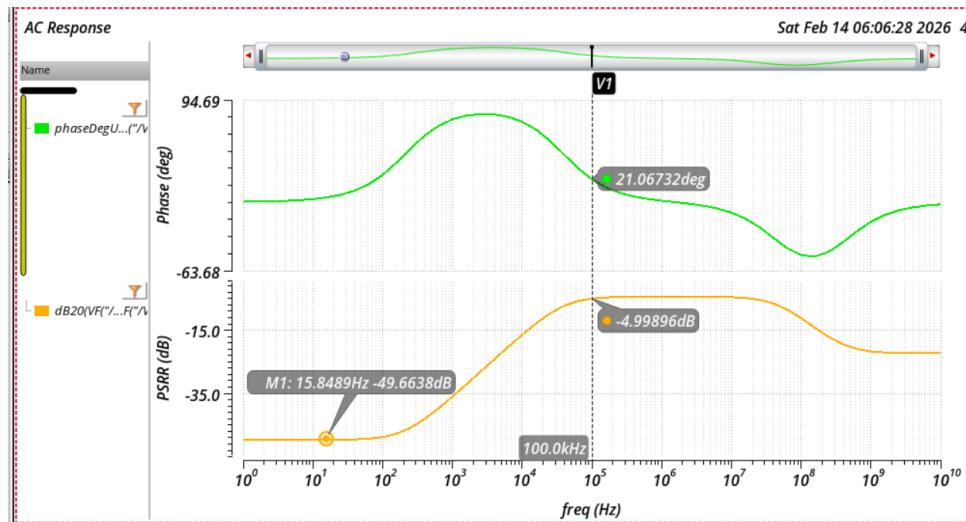


Figure 9: PSRR of pmos LDO

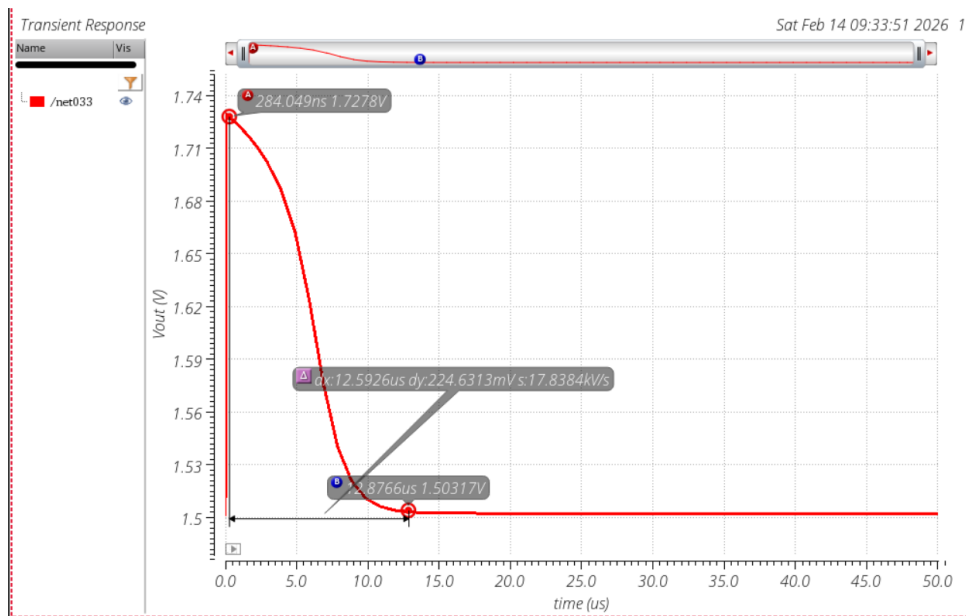


Figure 10: Overshoot of pmos LDO